

Restock a First Aid Kit SOP (1x Episode: 12 supplies)

One episode fills one first aid kit. All twelve supplies go in during that one episode.

The work has three parts. It starts with sorting. Eleven loose supplies go into the three compartments, and each one is marked off on the printed list as soon as it is seated. Then the roll gets a peg, goes in on the roll space in compartment 1, and is marked off the same way. Then the lid closes and both clips go shut.

The episode starts with the kit box open and empty and every check box on the list empty. It ends with the box clipped shut and no check box left empty.

One supply, one mark. Put one supply in its compartment. Then mark its line on the list with the marker. Only then take the next supply. The list is never marked in a batch at the end.

The supplies are all alike. Nothing on the tray has to be told apart from anything else, and no compartment is printed with what belongs in it. What decides where a supply goes is the **fill order** and nothing else.

The **fill order** never changes. **Compartment 1**, the big one on the right, is filled first. Then **compartment 2**, the small one at the front left. Then **compartment 3**, the one behind compartment 2 at the back left. Nothing is counted at any point. A compartment is filled until there is no clear floor left to lay another supply flat, and only then is the next one started. The roll goes in last of all, on the **roll space** that is kept clear for it in compartment 1.

Three rules never bend. Do not take the next supply before the last one has been marked off. Do not mark a line before its supply is seated in the box. Do not bring the lid over while a box is still unmarked or anything still has to go in.

The right gripper does the main work. It takes every supply off the tray. It puts every supply in the box. It takes the marker and marks every line on the list. It pegs the roll, takes it from the left gripper, and seats it. It brings the lid over and pushes both clips shut.

The left gripper helps. It holds the kit box down through the sorting, the marking, the seating of the roll, and the lid swing. It comes off the box for one job only: it holds the roll up while the peg goes on. It presses the lid down while the clips go shut. It never touches the marker or the list, and it never carries anything to the kit box.

The kit box is never lifted or carried. It stays where it is set up. All the work happens there.

Setup

Go through both checklists before you start an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual
- **parked_side:** none

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table. It shows the kit box at the center, the supply tray at the front center right in front of the kit box, the printed list and the marker park at the front left, and the reject bin at the right edge.
3. The kit box can be seen from above. All three compartments and the bottom of each one are in view with the lid folded back, compartment 3 at the back left included, and any bare floor left in a compartment can be seen from the footage.
4. The printed list can be seen from above. All twelve lines and all twelve check boxes are in view, and a mark going into a box can be seen as it is drawn. The marker park is in view beside it.
5. The roll can be seen while the left gripper holds it up. The peg going onto its rim is in view, and so is the loose end of the bandage under the peg.
6. Both arms are at home with grippers open.
7. The table is clean, dry, and empty. The only things on it are the kit box, the supply tray, the printed list, the marker in its park, and the reject bin.
8. The right arm reaches the supply tray, all three compartments, the pegs, the roll while the left gripper holds it, the reject bin, the marker park, and all twelve check boxes on the list. It does this without leaning out and without hitting a joint limit.
9. The left arm reaches the left end of the kit box, the roll on the supply tray, and the top of the lid. It does this without crossing to the right of the supply tray.

Materials checklist

1. One **kit box** sits at the center of the table. It is square to the table with its long side facing the front edge. It is **open**, and the **lid** is folded back and lying flat behind it.
2. Inside the kit box is one **tray** with **three compartments**. They are not in a row. One big compartment takes up the whole right half of the tray, and the left half is split front and back into two smaller ones. Nothing is printed on the tray rim. They are told apart by where they sit, and this SOP numbers them:
 - **Compartment 1:** the big one on the **right**. It runs the whole depth of the tray. It is filled first, and the **roll space** at its back end is kept clear for the pegged roll.
 - **Compartment 2:** the small one at the **front left**, nearest the front rail. It is filled second.
 - **Compartment 3:** the one **behind compartment 2**, at the **back left**. It is filled last.

3. Every compartment is empty, dry, and clean before the episode starts.
4. The eleven supplies and the roll fill the three compartments exactly. Every one of them lies flat, side by side, with none on top of another and none sticking up above the tray rim. Nothing has to be counted during the episode: the compartments run out of floor at the same moment the supply tray runs out of supplies. If they do not, the station is staged wrong.
5. The **roll space** is marked on the floor of compartment 1, at its back end, with a printed outline the size of the roll. It is the only marking anywhere in the tray. No supply ever goes on it. Only the pegged roll goes there, at the end.
6. The wall between compartment 2 and compartment 3 runs left to right across the left half of the tray. Both of those compartments are open from above with the lid folded back, so a supply goes straight down into either one without being brought in over the other.
7. The kit box has two **clips** on its front rail. There is a **left clip** and a **right clip**. Both hang open and away from the lid lip.
8. The kit box is heavy enough that it does not slide when a supply goes into a compartment. The left gripper holds it anyway, so it cannot creep out of place over twelve puts.
9. Eleven loose **supplies** lie on the **supply tray**. The tray sits at the front center, right in front of the kit box, between the box and the front edge of the table. The supplies are all the same size and shape, and each one has a flat face to sit on. Any of them can go in any place the fill order calls for, so none of them has to be picked out from the others.
10. One **roll** lies **flat face down** on the **right hand end of the supply tray**, in its own clear space, well away from the eleven supplies. It is wound tight, and the **loose end** of the bandage lies flat against its rim where a peg can close over it. It is not stacked on anything and nothing is lying on it.
11. Two **pegs** lie in a row along the **left hand end of the supply tray**, in their own clear space. A peg is a small spring clip. Squeezing its **tail** opens its jaws. Letting the tail go closes them again. One peg goes on the roll. The other is a spare and is only used if the first one fails.
12. Each peg opens wide enough to go over the roll's rim and grips hard enough to hold the loose end down once it is closed. No peg is cracked, bent, or missing its spring.
13. One **printed list** lies flat at the front left, square to the table. It is a plain list of the twelve supplies that go in the kit: **twelve identical lines, one per supply**, running down the sheet. Each line has a printed **check box** at its **right end**, so the right gripper reaches it without leaning across the table. No line is named or numbered and no line says which compartment it belongs to, because every supply on the tray is the same. The lines are marked from the top down, one per supply, in the order the supplies go in the box.
14. The list is a **fresh sheet** every episode. Every one of its twelve check boxes is empty before the episode starts. A list with a mark already on it is never reused.
15. The list is **well seated**. It sits flat on the table on a non-slip backing, so the marker can be drawn across a check box without pushing the sheet away. It may shift a little under the tip.

That is fine as long as it stays flat, stays square enough to read, and no line or check box goes out of the camera's view. No gripper ever holds the list down.

16. One **marker** lies in the **marker park**, a shallow groove on the table just above the printed list, at the front left. The marker is uncapped, its tip points to the right, and it leaves a clear line without being pressed hard. It is the only thing that ever makes a mark on the list.
17. The **reject bin** sits at the right edge of the table, clear of the kit box. It is empty. It takes a peg that will not hold. Nothing else goes in it.

Workspace layout

- **Kit box:** center. Open, lid folded back. Compartment 1 fills the right half of the tray with the roll space outlined at its back end, compartment 2 is at the front left, compartment 3 sits behind it at the back left.
- **Supply tray:** front center, right in front of the kit box. The eleven loose supplies in the middle, the **roll** in its clear space at the right hand end, and the **two pegs** in a row at the left hand end.
- **Printed list:** front left. Flat, seated, and square to the table, twelve identical lines with an empty check box at the right end of each one.
- **Marker park:** front left, in the groove just above the printed list. The marker lies in it whenever it is not making a mark.
- **Reject bin:** right edge of the table.

Everything the right gripper carries or touches sits on the right or at the center, and the list's check boxes sit at the right end of the lines for the same reason. The left gripper works on the left end of the kit box and on the roll, and on nothing else. Neither gripper crosses to work in the other one's area.

Arm assignments

- **Right gripper:** takes each supply off the supply tray. Puts each supply in the compartment the fill order calls for. Takes the marker out of its park, marks the next line, and lays the marker back. Takes a peg off the tray and closes it on the roll. Takes the roll from the left gripper, seats it on the roll space in compartment 1, and marks the last line. Brings the lid over. Pushes both clips shut.
- **Left gripper:** holds the kit box down by its **left end** through the sorting, the marking, the seating of the roll, and the lid swing. It leaves the box once, to lift the **roll** off the supply tray and hold it still while the peg goes on, and it goes straight back to the box after the hand-over. Presses the **lid** down while the clips go shut. It never carries anything to the kit box, never seats anything, and never touches the marker or the list.

Vocabulary

- **Kit box:** the hard case at the center of the table. It holds the tray with the three compartments. Its **front rail** is the rim edge nearest the front edge of the table. The two clips are on it. Its **left end** is the short rim edge at its left side, clear of the front rail and of both clips. That is the only place the left gripper holds the box. The box is never lifted or carried.
- **Lid:** the top of the kit box. It is on a hinge. It starts **folded back**, lying flat behind the box, and it comes over once at the end. Its **lid lip** is the front edge that the clips grab.
- **Clip:** one of the two catches on the front rail. A **closed clip** has been pushed up over the lid lip and stays there when the gripper lets go. An **open clip** hangs down, away from the lid.
- **Compartment:** one walled space in the tray. There are three, and nothing is printed on the rim, so they are told apart by where they sit. **Compartment 1** is the big one on the right. **Compartment 2** is the small one at the front left. **Compartment 3** is the one behind it at the back left.
- **Fill order:** the order the compartments are filled, and the only thing that decides where a supply goes. **Compartment 1** first, then **compartment 2**, then **compartment 3**. Nothing is counted. A supply goes in the compartment the order has reached while that compartment still has **clear floor**, and the next compartment is started only once it has none. The pegged roll goes in last of all, on the roll space in compartment 1.
- **Clear floor:** bare compartment floor with room to lay one more supply flat, inside the walls, beside the ones already there and not on top of one. The roll space in compartment 1 is never clear floor. A compartment with no clear floor is full.
- **Roll space:** the outline printed on the floor of compartment 1 at its back end. It is the size of the roll and it stays bare until the pegged roll goes in at the end.
- **Supply:** one loose item on the supply tray. They are all alike. There are eleven of them, and the roll makes the twelfth thing that goes in the box.
- **Roll:** the one wound roll in its clear space at the right hand end of the supply tray. Its **loose end** is the tail of bandage lying against its rim. It goes in the box only after it has a peg on it, and it goes in last, in compartment 1.
- **Peg:** a small spring clip. There are two on the tray. Squeezing its **tail** opens its jaws, and letting the tail go closes them. One peg goes on the roll and stays on it. The other is a spare.
- **Pegged roll:** the roll with one peg closed on its rim over the loose end. The peg holds the roll wound. Nothing else is holding it.
- **Flat face down:** the supply or the roll sits on one whole flat face. Nothing props it up, and no part of it stands on its edge.
- **Seated supply:** a supply sitting on the bottom of its compartment, flat face down, inside the walls. All of it is below the tray rim. It does not sit on top of another supply.

- **Full compartment:** a compartment with no **clear floor** left. Every supply in it is seated flat, nothing sits on top of anything, and nothing sticks up above the tray rim. Compartment 1 counts as full once its floor is covered up to the roll space, and it is only finished when the pegged roll is on that space.
- **Hand-over:** the one moment both grippers are on the roll. The left gripper holds it still, the right gripper closes on it, and only then does the left gripper open. The roll is never let go by both at once and never dropped from one to the other.
- **Printed list:** the seated sheet at the front left. It is a plain list of the twelve supplies, twelve identical lines with a check box at the right end of each one.
- **Check box:** the printed box at the right end of one line. It starts empty and takes one mark.
- **Marker:** the pen that lives in the marker park. It is uncapped and leaves a clear line without being pressed hard. It is the only thing that ever marks the list, and it goes back in its park after every mark.
- **Marker park:** the shallow groove just above the printed list. The marker lies in it whenever it is not making a mark. The gripper is never carrying the marker and a supply at the same time.
- **Marker tip:** the writing end of the marker. It is what touches the paper. It comes down inside a check box and nowhere else on the sheet.
- **Mark a line:** the right gripper takes the **marker** out of its park by the barrel, brings the **marker tip** down inside that line's **check box**, draws **one stroke** across the box from corner to corner, lifts the tip straight up off the paper, carries the marker back, and lays it in the **marker park**. The mark stays on the sheet, and the camera sees it go on.
- **Marked box:** a check box with one stroke drawn through it. It is the record that its supply is in the kit.
- **Next line:** the line for the supply that has just been seated. It is always the next unmarked line down, because the lines are all the same and are worked from the top. The first line goes with the first supply into compartment 1, the twelfth with the pegged roll.
- **Held still:** the gripper stops moving and stays put. It does not wave, drift, or bounce.
- **Lift:** come straight down onto the grip point, close, and go straight up. Keep going up until there is daylight under the thing. Only then move sideways.
- **Release point:** over a compartment, over the supply tray, over the reject bin, in the marker park, or into the other gripper at the hand-over. A gripper opens nowhere else. The gripper never opens over the printed list.

Steps

Step 1 is one loose supply, seated and marked off. It runs eleven times, in **fill order**. Steps 2 and 3 are the roll. They run once. Steps 4 to 6 run once.

Step 1: Put one supply in its compartment and mark off its line

Goal: one supply sits in the compartment the **fill order** calls for, and its line on the list has one mark in its check box. The fill order is not broken.

1.1 Take one supply

- Before the first supply comes off the tray, the **left gripper** closes on the **left end** of the **kit box** and holds it down on the table. It takes the left end, not the front rail, so it is clear of the supply tray in front of the box and clear of both clips.
- Any supply on the tray will do. They are all alike. Take the one nearest the right gripper that is clear of the others, so the gripper does not have to dig one out. Leave the roll and the pegs alone.
- The **right gripper** closes on that **supply** on the **supply tray** in front of the kit box. It **lifts** the supply straight up off the tray.

Take one supply per grip. Never close on two at once. Never drag or slide a supply along the tray or the table. Never take a supply back out of a compartment. Never take the next supply while the last one is still unmarked on the list. Never pick a supply up while the marker is in the gripper.

Check: the right gripper holds one supply up off the tray. The roll and both pegs are still in their spaces on the tray. The marker is in its park. The last supply that went in the box has its box marked.

1.2 Put the supply in its compartment

- The **right gripper** carries the supply flat back over the **kit box**. Which compartment it goes over comes from the **fill order** and from nothing else. Nothing is counted. Look for **clear floor**.
 - While **compartment 1**, the big one on the right, still has clear floor, the supply goes over compartment 1.
 - Once compartment 1 has no clear floor left up to the roll space, the supply goes over **compartment 2**, the small one at the front left.
 - Once compartment 2 has no clear floor left, the supply goes over **compartment 3**, at the back left.
- Look into the compartment before you come down. If there is a bare patch of floor big enough to lay the supply flat beside the others, that compartment is still the one. If there is not, move on to the next one in the order.
- The **roll space** at the back end of compartment 1 is never clear floor. Its outline stays bare until the roll goes in at the end.
- Lower the supply into the compartment **flat face down**. Come straight down into the compartment. Never bring a supply in for compartment 2 across compartment 3, or the

other way round. Keep going down until it sits on the bottom, inside the walls, beside the ones already there. Then let go.

- The **left gripper** keeps holding the left end down the whole time.

Carry it all the way down before you let go. Never drop a supply in from above the tray rim. Never stand one on its edge. Never put one on top of another. Never squeeze a supply into a gap it does not lie flat in. Never start compartment 2 while compartment 1 still has clear floor, and never start compartment 3 while compartment 2 still has clear floor. Never lay a supply on the roll space.

Check: the supply is a **seated supply**. It lies flat face down on the bottom of the compartment. All of it is below the tray rim. It is not on top of another supply and not on the roll space. It is in the compartment the fill order had reached. If it stands on edge, sits on another supply, sticks up above the rim, went on the roll space, or went in the wrong compartment, the **right gripper** lifts it out and puts it in again **before** the line is marked.

Expected state: that compartment has one more seated supply in it. The kit box has not moved. The right gripper is empty. The line for that supply is still unmarked.

1.3 Mark off its line on the list

- The **right gripper** moves from the kit box to the **marker park** at the front left and closes on the **marker** by its barrel. It lifts the marker clear of the park.
- It brings the **marker tip** down inside the **next line's** check box, the next unmarked one down, and draws **one stroke** across the box from corner to corner. It presses only hard enough to leave a line.
- It lifts the marker straight up off the paper, carries it back to the **marker park**, lays it in the groove, and lets go.
- Only then does it go back for the next supply.
- The **left gripper** stays on the kit box left end the whole time. It never comes to the marker or the list.

One mark per supply, from the top down. Mark the line only after its supply is seated. Never draw more than one stroke in a box. Never let the tip touch the sheet outside a check box. Never carry the marker over the kit box. Never hold the marker and a supply at the same time. Never leave the marker out of its park while a supply is being picked or seated.

Check: one stroke sits inside the next unmarked box down and nowhere else on the sheet. The marker is back in its park. The list is still flat and readable at the front left. It may have shifted a little, but every line and every box can still be seen, and it is not folded, torn, or pushed off its place.

Retry: if the stroke lands outside the box, runs across two boxes, or comes out too faint to see, draw one more stroke inside the right box, then carry on. If the marker is dropped, pick it up, lay

it in its park, and take it again from there.

Expected state: one new mark has gone on the sheet for the supply just seated, and the marker is back in its park.

Go back to 1.1 with the next supply while any supply is still on the supply tray. Once the tray holds no supply, only the roll and the pegs, and the only empty check box left is the bottom one, go on to Step 2. If a compartment still has clear floor when the tray is empty, or a supply is left over when no compartment has clear floor, end the episode there and log a staging fault.

Step 2: Put a peg on the roll

Goal: one peg is closed on the roll's rim over the loose end, and the roll is held up off the tray the whole time.

2.1 Take up the roll

- The **left gripper** lets go of the kit box **left end** and moves to the **roll** in its space at the right hand end of the supply tray.
- It closes on the side of the roll, clear of the rim where the peg is going, and **lifts** the roll straight up off the tray until there is daylight under it.
- It holds the roll still, flat face down, out over the tray, with the **loose end** of the bandage facing up where the right gripper can reach it. It is **held still** there until the hand-over in 3.1.
- Look the roll over. It is wound tight. The loose end lies flat against the rim.

The left gripper lifts the roll and then stops moving. Never carry the roll to the kit box with the left gripper. Never start on the roll while a supply is still on the tray or more than one check box is still empty. Never work the roll while it is lying on the tray.

Check: the left gripper holds the roll up off the supply tray, still, flat face down, with the loose end in view. The kit box has not moved. The right gripper is empty and the marker is in its park.

Retry: if the roll has come unwound in the lift, the **left gripper** sets it back down flat face down in its space on the tray and takes it up again. If it will not stay wound in the hand, put it down and log a bad roll.

2.2 Put the peg on

- The **right gripper** closes on the **tail** of the nearer **peg** at the left hand end of the supply tray. That squeezes the peg open. It lifts the peg straight up off the tray.
- It brings the open peg over the roll's rim with the **loose end** under its jaws.
- It lets the tail go. The peg closes on the rim and holds the loose end down.
- The **right gripper** comes away from the peg and stops clear of the roll.
- The **left gripper** does not move while the peg goes on.

One peg per roll. Squeeze the tail to open it, then let go to close it. Never force the peg's jaws apart by any other means. Never put the peg anywhere but on the rim over the loose end. Never take the second peg while the first one is still good.

Check: the roll is a **pegged roll**. One peg is closed on its rim with the loose end under it. The roll is still wound tight. The peg holds when the right gripper lets go. The second peg is still in its space on the tray.

Retry: if the peg falls off, closes on nothing, or lets the roll unwind, the **right gripper** picks that peg up, carries it to the **reject bin**, and lets go over the bin. Then it takes the **spare peg** and 2.2 runs again. If the second peg will not hold either, the **left gripper** sets the roll back down on the tray, both arms come away, and the episode ends there with a bad peg logged.

Step 3: Put the pegged roll in the box and mark off the last line

Goal: the pegged roll sits on the roll space in compartment 1, and the last empty box on the list has been marked.

3.1 Hand the roll over and seat it

- The **right gripper** closes on the **pegged roll**, clear of the peg, while the **left gripper** is still holding it. That is the **hand-over**. Only once the right gripper has it does the left gripper open and come away.
- The **left gripper** goes straight back to the **left end** of the kit box and holds it down again. It stays there through Step 4.
- The **right gripper** carries the roll flat to **compartment 1** and holds it over the **roll space** at the back end, the outline that has been kept bare all episode.
- Lower it in **flat face down** until it sits on the bottom inside the outline, beside the supplies and not on top of one. The peg lies flat along the rim. Then let go.

Both grippers are on the roll before either lets go. Never toss or drop the roll from one gripper to the other. Never take the peg off to make the roll fit. Never seat the roll before the left gripper is back on the kit box. Never put it in compartment 2 or compartment 3.

Check: the roll is seated on the roll space in compartment 1. It lies flat face down on the bottom, inside the outline, beside the supplies and not on top of one. The peg is still closed on its rim over the loose end. All of it, the peg included, is below the tray rim. No compartment has clear floor left, and nothing is left on the supply tray but the spare peg.

Retry: if the roll sits on top of a supply, stands proud of the rim, or the peg comes off as it goes in, the **right gripper** lifts it back out. If the peg came off, hold the roll up in the **left gripper** and run 2.2 again with the spare peg. Then seat it again. The line stays unmarked until the roll is properly seated.

3.2 Mark off the last line

- The **right gripper** takes the **marker** out of its park, brings the **marker tip** down inside the bottom line's check box, the last unmarked one, and draws **one stroke** across it from corner to corner.
- It lifts the marker straight up, carries it back to the **marker park**, lays it in the groove, and lets go.
- The **left gripper** stays on the kit box left end.

Check: no check box on the list is left empty, and no box carries more than one mark. The marker is back in its park. The list is still flat and readable at the front left.

Retry: if the stroke lands outside the box or comes out too faint to see, draw one more stroke inside the right box.

Expected state: every compartment full, the roll on its space in compartment 1, no empty box left on the list, and the marker in its park.

Step 4: Close the lid

Goal: the lid lies down on the kit box. The lid lip is over the front rail. Nothing is caught under it.

- The **left gripper** keeps holding the **left end** of the kit box down on the table for the whole step, so the lid swing cannot shove the box. The left end is clear of the front rail, so the lid lip can still come down onto the rail.
- Look in the tray one more time. No compartment has clear floor left, the pegged roll is on its space in compartment 1, and nothing sticks up above the tray rim. Nothing else is lying in the box.
- Look at the list one more time. No check box is left empty.
- The **right gripper** closes on the **lid** at the middle of its back edge. It swings the lid up and over the kit box in one move. It lowers the lid until it lies down on the tray rim. Then it lets go.
- Lower the lid the last bit of the way. Do not let it fall.

Never close the lid while a supply or a peg is still lying loose in the box. Never close it while any check box on the list is still empty. Never swing the lid with the marker in the gripper.

Check: the lid lies down all the way round. The **lid lip** sits over the **front rail**. No supply is caught between the lid and the rim, and the peg has not been squeezed off the roll. The box has not moved.

Retry: if the lid sits up on one side, or a supply is caught under it, the **right gripper** swings the lid back open while the **left gripper** keeps holding the left end. Put the caught supply back in as in 1.2. Its box stays marked. Then Step 4 runs again.

Step 5: Clip the lid shut

Goal: both clips are shut over the lid lip. Both hold on their own.

- The **left gripper** lets go of the left end, moves onto the middle of the **lid**, and presses it down onto the kit box. It stays there for the whole step, so the clips have something to push against.
- The **right gripper** closes on the **left clip**. It pushes the clip straight up onto the **lid lip** until it catches. Then it lets go.
- The **right gripper** then closes on the **right clip** and shuts it the same way.
- Shut the left clip first and the right clip second. Do not start the right clip before the left one has caught.
- The **left gripper** lets go of the lid once both clips are shut.

Push each clip straight up onto the lip. Reach over the supply tray to the front rail. Never push the supply tray out of place to get at a clip. Never force a clip when the lid is sitting up on one side. Never lift the kit box by a clip or by the lid.

Check: both clips are **closed clips**. Each one sits over the lid lip and stays there once the grippers are off. The lid lies down all the way round. The kit box is still square to the table, where it was set up, and so is the supply tray in front of it.

Retry: if a clip will not catch, or springs back open, the **right gripper** lets it hang open. The **left gripper** presses the lid down again. Push that clip up one more time. If it still will not catch after that second try, leave it open, finish the other clip, and log the box for a station check.

Step 6: End the episode

Goal: the kit box is shut and clipped, no check box on the list is left empty, the supply tray is cleared, the marker is back in its park, and both arms are safely home.

- Confirm the kit box is shut with both clips closed over the lid lip, square to the table where it was set up.
- Confirm no check box on the printed list is left empty and none carries more than one mark, and the list lies flat and readable at the front left with no mark anywhere off a box.
- Confirm the marker lies in its park and the supply tray holds no loose supply and no roll, with the spare peg still in its space at the left hand end.
- Confirm the reject bin holds every rejected peg and nothing else.
- Return both arms home, clear of the kit box and the list, with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Push both clips down open. Fold the lid back so it lies flat behind the kit box.
2. Take the pegged roll off the roll space in compartment 1. Squeeze the peg's tail, work it off the rim, and put it back in its space at the left hand end of the supply tray. Wind the roll up again if it has come loose, lay the loose end flat against the rim, and put the roll back flat face down in its space at the right hand end of the tray.
3. Take the loose supplies out of the three compartments. Lay them on the supply tray in front of the kit box, clear of the roll and of the pegs. Check the roll space outline in compartment 1 is bare and can still be seen.
4. Take any rejected peg out of the reject bin. Put it to one side. Do not put it back on the tray.
5. Make sure there are eleven supplies on the tray and two good pegs in the row at the left hand end.
6. Throw the marked list away. Put a **fresh printed list** down flat and square at the front left, on its non-slip backing, with all twelve check boxes empty. Check every line and every box can be seen from above.
7. Lay the marker back in the marker park above the list. Check it still leaves a clear line without being pressed hard.
8. Wipe the tray and the table. The surface is clean, dry, and empty apart from the kit box, the supply tray, the printed list, the marker in its park, and the reject bin, each in its own place.
9. Replace anything that is no good. That means a roll that will not stay wound with a peg on it. A peg that is cracked, bent, or has lost its spring. A marker that has dried out or leaves a line too faint to see. A kit box with a clip that will not catch.

SOP violations

These are actions that break the SOP. They are reviewed side by side in the review tool.

How to record a violation in review

For each violation, write down the **start timestamp**, the **violation name**, and the **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining. It is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag on it. Do not delete it just because a rule was broken. One episode covers a whole kit, so tag each violation at the time of the supply, the roll, or the mark it happened on.

Violations

Violation: Kit box not held down

- **Visible cue:** the left gripper is open, off the left end, or away from the kit box while the right gripper puts in a supply, marks a line, seats the roll, or swings the lid over. Or it never takes the left end at all. Or it takes the front rail instead and blocks the clips. Or it is still off the box at the hand-over and the roll is seated with the box unheld.
- **SOP rule broken:** Steps 1, 3.1, and 4, the left gripper presses the kit box left end down through the sorting and the marking, and goes straight back to it after the hand-over, before the roll is seated.
- **Coaching note:** the left gripper is the clamp. It comes off the box for the roll and for nothing else, and it is back on before the roll goes in.

Violation: Kit box moved or lifted

- **Visible cue:** either gripper lifts the kit box off the table, carries it, turns it, or slides it somewhere else. Or the box gets dragged along by a clip or by the lid.
- **SOP rule broken:** Steps 1, 4, and 5, the kit box stays where it is staged and is worked on in place, and it is never lifted by a clip or by the lid.
- **Coaching note:** bring the supply to the box. The box does not move.

Violation: Arm jobs swapped or one arm idle

- **Visible cue:** the left gripper takes a supply, carries the roll to the kit box, seats anything, takes the marker, marks a line, swings the lid, or pushes a clip. Or the right gripper holds the left end or the lid down instead of doing the work, or holds the roll still while the left gripper pegs it. Or one arm never leaves home while the other does the pick, the hold, and the put on its own.
- **SOP rule broken:** Steps 1 to 5, the right gripper picks, seats, marks, pegs, closes, and clips, and the left gripper holds the left end, the roll, and the lid.
- **Coaching note:** the left gripper holds. The right gripper works. The only thing the left gripper ever picks up is the roll, and it never carries it to the box.

Violation: Wrong pickup

- **Visible cue:** the right gripper closes on two or more supplies at once. Or it drags or slides a supply along the tray or the table instead of lifting it up. Or it takes a supply back out of a compartment. Or it picks up the roll or a peg during the sorting. Or it takes something out of the reject bin.
- **SOP rule broken:** Step 1.1, the right gripper lifts one supply at a time clear of the supply tray and leaves the roll and the pegs alone until Step 2.
- **Coaching note:** one supply per grip. The roll and the pegs have their own spaces on the tray and their own step.

Violation: Fill order broken

- **Visible cue:** a supply goes in compartment 2 while compartment 1 still has bare floor outside the roll space, or in compartment 3 while compartment 2 still has bare floor. Or the roll is taken up or pegged while a supply is still on the tray or more than one check box is empty. Or the compartments are filled a bit at a time, turn and turn about, instead of one after the other.
- **SOP rule broken:** Step 1.2 and the fill order, compartment 1 first, then compartment 2, then compartment 3, with the pegged roll last of all, and a compartment is started only once the one before it has no clear floor left.
- **Coaching note:** look at the floor, do not count. While there is a bare patch big enough to lay one flat, the supply goes there.

Violation: Supply in the wrong compartment

- **Visible cue:** a supply is forced into a compartment with no room left, so it sits on another one, leans on a wall, or stands proud of the rim. Or a supply is laid on the **roll space** outline in compartment 1. Or a supply for compartment 2 is brought in across compartment 3, or one for compartment 3 across compartment 2, instead of coming straight down into its own, and something gets knocked out of the compartment it crossed. Or a supply is left in a compartment the fill order had not reached instead of being lifted out and put in again.
- **SOP rule broken:** Steps 1.2 and 3.1, look for clear floor before coming down, keep the roll space bare until the end, come straight down into the compartment the fill order has reached clear of the one in front of or behind it, and reseat a supply that went in the wrong place before its line is marked.
- **Coaching note:** if it does not lie flat on bare floor, it does not belong there. The outlined space in compartment 1 is the roll's and stays bare.

Violation: Supply not seated

- **Visible cue:** a supply is let go above the tray rim and drops in. Or it lands on its edge, sits on top of another supply, sticks up above the rim, or leans on a compartment wall. Or it is left like that and its line is marked anyway.
- **SOP rule broken:** Step 1.2, lower the supply flat face down until it rests on the compartment floor inside the walls, then release, and reseat it before the line is marked.
- **Coaching note:** carry it down to the bottom of the compartment before you open the gripper. Fix the seating first, mark the line second.

Violation: Mark skipped or out of step

- **Visible cue:** a supply is seated and the right gripper goes straight back to the tray for the next one without marking a box. Or several supplies go in and then several boxes get marked in a row to catch up. Or a box is marked before its supply is seated. Or the episode ends with empty boxes left on a list that should be full, or with marks on a list while supplies are still on the tray.

- **SOP rule broken:** Step 1.3 and Step 3.2, one supply then one mark, the box marked only after its supply is seated, and never a batch of marks.
- **Coaching note:** seat it, then mark its line, then take the next one. The list is kept in step with the box, not caught up at the end.

Violation: Wrong line marked

- **Visible cue:** the stroke goes in a box that is not the next unmarked one down. Or boxes are marked out of order instead of from the top down. Or a box that already carries a mark is marked again.
- **SOP rule broken:** Step 1.3, mark the **next line**, one box at a time, from the top of the list down.
- **Coaching note:** the lines are all the same, so the right one is always the next one down. One box per supply.

Violation: Mark made wrong

- **Visible cue:** the stroke lands outside the box, runs across two boxes, or crosses the printed line instead of sitting in the box. Or more than one stroke goes in a box. Or the marker is scribbled, or pressed so hard it folds, tears, or shoves the list. Or the mark comes out too faint to see and is left that way.
- **SOP rule broken:** Step 1.3, bring the marker tip down inside the check box, draw one stroke across it corner to corner, and press only hard enough to leave a line.
- **Coaching note:** tip down inside the box, one stroke, lift. If it comes out faint, one more stroke in the same box and move on.

Violation: Marker handled wrong

- **Visible cue:** the marker is left out of its park while a supply is picked or seated. Or it is carried over the kit box, or held at the same time as a supply, or put down anywhere but the park. Or it is gripped by the tip instead of the barrel. Or it is dropped and left where it fell. Or the left gripper takes it.
- **SOP rule broken:** Step 1.3, the right gripper takes the marker out of its park by the barrel and lays it back in the park after every mark, and it is never carried with a supply or over the kit box.
- **Coaching note:** the marker comes out for one stroke and goes straight back. Nothing else is in the gripper while it is out.

Violation: List knocked out of place

- **Visible cue:** the list is dragged, folded, torn, flipped, or pushed off the front left so a line or a check box can no longer be seen from above. Or a gripper grabs the list, or opens over it, or carries a supply across it.

- **SOP rule broken:** Materials checklist item 15 and Step 1.3, the list is well seated and may shift a little, but it stays flat, readable, and in its place, and no gripper grips it or opens over it.
- **Coaching note:** the tip rests inside the box. It does not push, shove, or grab. A little shift is fine. A list you cannot read is not.

Violation: Roll held or handed over wrong

- **Visible cue:** the peg goes on while the roll is still lying on the tray. Or the left gripper waves, drifts, or moves the roll about while the peg goes on. Or it grips the roll over the rim where the peg has to go. Or the left gripper carries the roll to the kit box, or lets go before the right gripper has it, or the roll is tossed or dropped between the two grippers. Or both grippers let go at once.
- **SOP rule broken:** Steps 2.1 and 3.1, the left gripper lifts the roll clear of the tray and holds it still, clear of the rim, and at the hand-over the right gripper closes on the roll before the left one opens.
- **Coaching note:** lift it, hold it dead still, and let the right gripper take it off you. Two grippers on the roll before either lets go.

Violation: Peg taken or handled wrong

- **Visible cue:** the right gripper closes on a peg anywhere but its tail, so the peg does not open. Or it takes both pegs, or takes the spare while the first peg is still good. Or it prises the jaws open some other way. Or it drags a peg along the tray instead of lifting it. Or the left gripper takes the peg.
- **SOP rule broken:** Step 2.2, the right gripper closes on the peg's tail to squeeze it open, lifts it clear of the tray, and takes the spare only after the first peg has gone to the reject bin.
- **Coaching note:** grip the tail, not the jaws. That is what opens it. One peg at a time.

Violation: Peg not on the roll properly

- **Visible cue:** the peg closes on nothing, on the roll away from the loose end, or over the left gripper's jaws. Or it falls off when the right gripper lets go and the roll goes in the box anyway. Or the roll is clearly unwound under the peg. Or a peg that failed is left on the tray or on the table instead of going to the reject bin.
- **SOP rule broken:** Step 2.2, one peg closes on the roll's rim over the loose end and holds when the right gripper lets go, and a peg that fails goes to the reject bin before the spare is taken.
- **Coaching note:** the loose end goes under the jaws. Let go and watch it hold before you move on.

Violation: Roll with no peg goes in the box

- **Visible cue:** the roll goes from the supply tray or the left gripper into compartment 1 with no peg on it. Or the peg is taken off to make the roll fit. Or someone works at the roll once it is already in a compartment as if pegging it there.
- **SOP rule broken:** Steps 2 and 3.1, the roll is pegged while the left gripper holds it up, and it keeps its peg once it is in the box.
- **Coaching note:** peg it in the air, then seat it, then mark the last line. Nothing is pegged inside the box.

Violation: Pegged roll not seated

- **Visible cue:** the pegged roll is let go above the tray rim, lands on its edge, sits on top of a supply, sticks up above the rim with the peg proud of it, goes anywhere in compartment 1 but its outlined space, or goes in compartment 2 or compartment 3.
- **SOP rule broken:** Step 3.1, lower the pegged roll flat face down onto the roll space in compartment 1 until it rests on the floor inside the outline, then release.
- **Coaching note:** the same as a supply. Down to the bottom, flat, inside the outline, peg lying along the rim, then let go.

Violation: Fixture knocked out of place

- **Visible cue:** a gripper grabs, pushes, drags, or knocks the supply tray, the marker park, or the reject bin out of its place. Or the roll or the pegs get shoved out of their spaces on the tray while supplies are being picked. Or the supply tray in front of the kit box gets shoved while an arm reaches the clips or the list.
- **SOP rule broken:** Steps 1.1, 1.3, 2.1, 2.2, and 5, the tray, the bin, the marker park, the roll's space, and the pegs' space all stay where they are staged, and no gripper grips or nudges them.
- **Coaching note:** the fixtures stay put. The supply tray sits right in front of the box, so lift the arm over it instead of brushing through it.

Violation: Lid closed on something

- **Visible cue:** the lid comes over while a supply or the roll sticks up above the tray rim, or while a peg, the marker, or a gripper is in the box. Or a supply is clearly caught between the lid and the rim and is left there. Or the lid squeezes the peg off the roll and it is left loose in the box.
- **SOP rule broken:** Step 4, look into the tray before the lid swings over, and reopen the lid and reseat a caught supply rather than leaving it trapped.
- **Coaching note:** one look in the tray before the lid moves. If something is caught, open it back up.

Violation: Lid closed with the list unfinished

- **Visible cue:** the lid comes over while a check box on the list is still empty. Or the boxes that were missed get marked after the lid is down or after the clips are shut.
- **SOP rule broken:** Step 4, close the lid only once no check box on the list is left empty.
- **Coaching note:** the list is finished when the box is finished. Run your eye down the column for an empty box before the lid moves. Nothing gets marked after the lid is down.

Violation: Lid closed wrong

- **Visible cue:** the lid is dropped or allowed to fall the last bit of the way. Or the left gripper swings it over. Or it is pushed shut from the side. Or it is left sitting up on one side and clipped anyway. Or it is swung with the marker still in the gripper.
- **SOP rule broken:** Step 4, the right gripper takes the lid at the middle of its back edge, swings it over in one motion, and lowers it down onto the tray rim.
- **Coaching note:** take the back edge, swing it over, set it down. Never let it fall.

Violation: Clips not shut properly

- **Visible cue:** the episode ends with a clip still hanging open. Or both clips are pushed at once. Or the right clip is shut before the left one has caught. Or a clip is forced while the lid sits up on one side. Or a clip is worked at more than twice instead of being left open and logged. Or a clip springs back open and is left with nothing holding the lid.
- **SOP rule broken:** Step 5, the left gripper presses the lid down while the right gripper closes the left clip first and the right clip second, with two tries at most on a clip that will not catch.
- **Coaching note:** press the lid. Then left clip. Then right clip. Two tries, then leave it and log it.

Violation: Something dropped or knocked over

- **Visible cue:** a supply, the roll, a peg, or the marker is dropped on the table or the floor. Or it is knocked off the supply tray or out of a compartment. Or it has to be chased and grabbed again. Or a gripper opens somewhere that is not a release point, the printed list included.
- **SOP rule broken:** Steps 1 to 3, each thing is lifted clear, carried level, and released only over a compartment, over the supply tray, over the reject bin, in the marker park, or into the other gripper at the hand-over.
- **Coaching note:** grip it, lift it up, carry it flat, and open the gripper only over the place it belongs. The marker goes back in its park every time.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a supply, the roll, a peg, or the marker still in a gripper or on the table. Or with a clip open. Or with an empty check box left on the list. Or with the list or the supply tray pushed out of place. Or with an arm away from home.
- **SOP rule broken:** Step 6, confirm the clipped kit box, no empty check box on the list, the list flat at the front left, the marker in its park, and the cleared supply tray; return both arms

home clear of the scene; then stop recording.

- **Coaching note:** check first. Homing is the last thing the arms do.

Non-violation failures

These failures are not caused by how the task was run. Log them as system issues. Throw the episode away. Never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Bad peg:** a peg that will not open when its tail is squeezed, that has lost its spring, or that will not hold on a roll sitting properly in its jaws, twice over. Replace both pegs before the next episode.
- **Bad roll:** a roll that will not stay wound even with a good peg closed over its loose end. Replace it before the next episode.
- **Bad marker:** a marker that has dried out, that leaves a line too faint to see from above, or that bleeds through the sheet. Replace it before the next episode.
- **Bad printed list:** a list that is creased, torn, or printed so faintly that a line or a check box cannot be seen from above. Or one that came out of the stack with a box already marked. Or a backing that has stopped gripping, so the sheet slides out from under the marker. Put a fresh list down before the next episode.
- **Bad kit box:** a clip that will not catch on a lid that lies down all the way round, or a lid that will not lie down on an empty tray. Or a compartment that will not take its share lying flat. Replace the box before the next episode.
- **Staging fault:** a compartment still has clear floor when the supply tray is empty, or a supply is left over when no compartment has clear floor. Or the roll will not lie inside its outlined space. Or there is no roll on the tray, fewer than two pegs, or no marker in the park. Set the station up again before the next episode.
- **Fixture moved by a rig fault:** the kit box, the supply tray, the marker park, the reject bin, or the printed list is knocked out of place by the rig and not by a gripper. Put the fixtures back before the next episode.

Annotation subtasks (from SOP)

1. Take hold of the kit box left end and take one supply off the supply tray
2. Put one supply in the compartment the fill order calls for
3. Take the marker out of its park
4. Draw one stroke in the next check box on the printed list
5. Lay the marker back in its park

6. Repeat until no compartment has clear floor and the supply tray is empty of supplies
7. Lift the roll off the supply tray and hold it still
8. Take a peg off the tray and close it on the roll's rim
9. Hand the roll to the right gripper and take hold of the kit box again
10. Put the pegged roll on the roll space in compartment 1
11. Mark the last check box on the printed list
12. Bring the lid over and close it
13. Push the left clip and the right clip shut
14. Confirm the end state, return both arms home, and end the episode